You said:

Ok! Lets use the the continued chat for some checks first.

Features that work right now:

importing the map
color is applied to all relevant faces.
light preview.
... probalbly a lot more
Just give me a list of what is done, what is buggy and what has jet to be implemented. No long explanations. Just pure overview.

ChatGPT said:

You said:

Ok! Im gonna be away some hours in about 15 minutes. If you need some info i can provide it befoire i leave. Then you will have have a lot of time. Try to get rid of the bugs. remember you have access to the source of ericw, the map, blender and trenchbroom if you need it. Everything is open source here.

Worked for 1m 3s

ChatGPT said:

You said:

Here come all your requests.
I connected a github with you. you shold be authorized to work there now.
https://github.com/ChnuckiErdbeer/Quake-Surface-Studio
your original extracted qss v0.11.0 zip
a folder containing the blenderfile i used
a folder containing an updated map file (a simple cube mao with a pillow in the middle, one info_player_start and 8 lights all 16mu from the wall, each has "light" "50", one is red, one is blue one is green and the five others each have a different "delay" value.

go for it
Here is the output of the compile chain and running the map in ironwail:
Using working directory '/mnt/sde1/FPS/IdTech2/QuakeDev/##mapping/blender_lightmap_tests'

Exporting map file '/mnt/sde1/FPS/IdTech2/QuakeDev/##mapping/blender_lightmap_tests/test02_compile.map'

Copying to '/mnt/sde1/FPS/IdTech2/QuakeDev/id1/maps/': /mnt/sde1/FPS/IdTech2/QuakeDev/##mapping/blender_lightmap_tests/test02_compile.map

Executing '/mnt/sde1/FPS/IdTech2/QuakeDev/##tools/ericw-tools-2.0.0-alpha11-Linux/qbsp -litwater /mnt/sde1/FPS/IdTech2/QuakeDev/id1/maps/test02_compile.map'

---- qbsp / ericw-tools 2.0.0-alpha11 ----

low priority not compiled into this version

logging to /mnt/sde1/FPS/IdTech2/QuakeDev/id1/maps/test02_compile.log
(truncate)

--- Options Summary ---

"litwater" was set to "1" (from command line)

Input file: /mnt/sde1/FPS/IdTech2/QuakeDev/id1/maps/test02_compile.map

Output file: /mnt/sde1/FPS/IdTech2/QuakeDev/id1/maps/test02_compile.bsp

INFO: using built-in palette.

---- LoadMapFile ----

12 entities

5 unique textures

21 texinfos

32 unique planes

---- ProcessMapBrushes ----

INFO: world extents calculated to 1352 units

[100%] time elapsed: 00:00:00.000109

7 brushes

42 sides

0 texture frames added

WARNING: unable to find texture skip

Processing hull 0...

---- Brush_LoadEntity ----

[100%] time elapsed: 00:00:00.000107

---- CountBrushes ----

7 SOLID brushes

7 brushes total

42 visible sides

INFO: calculating BSP for 7 brushes with 42 sides

---- BrushBSP ----

7 brushes

42 visible faces

[done] time elapsed: 00:00:00.000449

24 nodes

24 expensive split nodes

25 leaves

---- CountLeafs ----

7 SOLID leafs

18 EMPTY leafs

25 leafs total

12 max tree height

7 avg tree height

---- MakeTreePortals ----

[100%] time elapsed: 00:00:00.000226

---- CalcTreeBounds ----

[done] time elapsed: 00:00:00.000026

118 tree portals

---- FillOutside ----

14 outside leaves

[done] time elapsed: 00:00:00.000019

---- FillDetail ----

0 detail filled leafs

---- BrushBSP ----

7 brushes

10 visible faces

32 non-visible faces

[done] time elapsed: 00:00:00.000319

28 nodes

18 non-visible nodes

28 expensive split nodes

29 leaves

---- CountLeafs ----

7 SOLID leafs

22 EMPTY leafs

29 leafs total

11 max tree height

6 avg tree height

---- MakeTreePortals ----

[100%] time elapsed: 00:00:00.000193

---- CalcTreeBounds ----

[done] time elapsed: 00:00:00.000022

134 tree portals

---- FillOutside ----

18 outside leaves

[done] time elapsed: 00:00:00.000015

---- FillDetail ----

0 detail filled leafs

---- MakeTreePortals ----

[100%] time elapsed: 00:00:00.000134

---- CalcTreeBounds ----

[done] time elapsed: 00:00:00.000018

134 tree portals

---- MarkVisibleSides ----

40 sides visible

---- MakeFaces ----

20 makefaces

4 merged

16 subdivided

---- PruneNodes ----

18 nodes pruned

---- WritePortalFile ----

[done] time elapsed: 00:00:00.000095

4 player-occupiable leaves

4 clusters of leaves

4 vis portals

---- CountLeafs ----

4 EMPTY leafs

7 SOLID leafs

11 leafs total

11 max tree height

6 avg tree height

---- TJunc ----

[100%] time elapsed: 00:00:00.000167

28 edges added by tjunctions

32 faces optimized through MWT

92 triangles computed from MWT

8 faces created via optimization of MWT triangles

---- EmitFaces ----

86 edges

40 faces

Processing hull 1...

Processing hull 2...0

---- FinishBSPFile ----

Writing /mnt/sde1/FPS/IdTech2/QuakeDev/id1/maps/test02_compile.bsp as Quake BSP bsp29

Wrote /mnt/sde1/FPS/IdTech2/QuakeDev/id1/maps/test02_compile.bsp

count	lump name	byte size
1	models	64
29	planes	580
46	vertexes	552
10	nodes	240
9	texinfos	360
40	faces	800
26	clipnodes	208
5	leafs	140

```
    42 marksurfaces        84
    87 edges              348
   172 surfedges          688
     5 textures         21944
       lightdata            0
       visdata              0
       entdata            947
```

0.017s seconds elapsed

Finished with exit code 0

Executing '/mnt/sde1/FPS/IdTech2/QuakeDev/##tools/ericw-tools-2.0.0-alpha11-Linux/vis /mnt/sde1/FPS/IdTech2/QuakeDev/id1/maps/test02_compile'

---- vis / ericw-tools 2.0.0-alpha11 ----

low priority not compiled into this version

logging to /mnt/sde1/FPS/IdTech2/QuakeDev/id1/maps/test02_compile-vis.log (truncate)

--- Options Summary ---

LoadBSPFile: '/mnt/sde1/FPS/IdTech2/QuakeDev/id1/maps/test02_compile.bsp'

BSP is version Quake BSP bsp29

INFO: using built-in palette.

Loading extended texinfo flags from /mnt/sde1/FPS/IdTech2/QuakeDev/id1/maps/test02_compile.texinfo.json...

4 leafs

4 clusters

4 portals

Calculating Base Vis:

[100%] time elapsed: 00:00:00.000015

Calculating Full Vis:

[100%] time elapsed: 00:00:00.000014

Expanding clusters...

average leafs visible: 3

c_noclip: 0

c_chains: 12

visdatasize:4 compressed from 4

---- CalcAmbientSounds ----

[100%] time elapsed: 00:00:00.000010

Writing /mnt/sde1/FPS/IdTech2/QuakeDev/id1/maps/test02_compile.bsp as Quake BSP bsp29

0.00s elapsed

Finished with exit code 0

Executing '/mnt/sde1/FPS/IdTech2/QuakeDev/##tools/ericw-tools-2.0.0-alpha11-Linux/light /mnt/sde1/FPS/IdTech2/QuakeDev/id1/maps/test02_compile'

---- ClearLightmapSurfaces ----

---- light / ericw-tools 2.0.0-alpha11 ----

low priority not compiled into this version

logging to /mnt/sde1/FPS/IdTech2/QuakeDev/id1/maps/test02_compile-light.log (truncate)

LoadBSPFile: '/mnt/sde1/FPS/IdTech2/QuakeDev/id1/maps/test02_compile.bsp'

BSP is version Quake BSP bsp29

INFO: using built-in palette.

---- load_textures ----

load: WARNING: filesystem error: cannot get file size: Is a directory [/mnt/sde1/FPS/IdTech2/QuakeDev/id1/maps/]

ConvertTextures: WARNING: invalid size data for

Loading extended texinfo flags from /mnt/sde1/FPS/IdTech2/QuakeDev/id1/maps/test02_compile.texinfo.json...

Loading extended content flags from /mnt/sde1/FPS/IdTech2/QuakeDev/id1/maps/test02_compile.content.json...

---- LoadEntities ----

12 entities read, 10 are lights.

--- Options Summary ---

"bounce" was set to "2" (from map)

"bouncescale" was set to "1.000000" (from map)

"bouncecolorscale" was set to "1.000000" (from map)

Embree_TraceInit: Embree version: 4.4.0

Embree_TraceInit:

0 sky faces

40 solid faces

0 filtered faces

0 shadow-casting skip faces

SetupLights: 10 initial lights

---- MakeSurfaceLights ----

SetupLights: 10 after surface lights

SetupLights: 10 after jittering

Final count: 10 lights, 0 suns in use.

---- LightWorld ----

---- ClearLightmapSurfaces ----

---- CalculateVertexNormals ----

[100%] time elapsed: 00:00:00.000117

---- MakeFaceCache ----

[100%] time elapsed: 00:00:00.000118

---- CreateLightmapSurfaces ----

[100%] time elapsed: 00:00:00.000169

---- MakeRadiositySurfaceLights ----

[100%] time elapsed: 00:00:00.000006

0 surface light points in use.

---- Direct Lighting ----

[100%] time elapsed: 00:00:00.000410

---- MakeBounceLights ----

[100%] time elapsed: 00:00:00.000061

---- Indirect Lighting (pass 0) ----

[100%] time elapsed: 00:00:00.001636

---- MakeBounceLights ----

[100%] time elapsed: 00:00:00.000028

---- Indirect Lighting (pass 1) ----

[100%] time elapsed: 00:00:00.001596

---- Post-Processing ----

[100%] time elapsed: 00:00:00.000020

---- SaveLightmapSurfaces ----

[100%] time elapsed: 00:00:00.000026

map uses color: true

Colored light entities/settings detected: .lit output enabled.

lightmap size (total): 15696

[100%] time elapsed: 00:00:00.000033

Lighting Completed.

Writing /mnt/sde1/FPS/IdTech2/QuakeDev/id1/maps/test02_compile.lit

---- ClearLightmapSurfaces ----

0 switchable light styles (32 max)

Writing /mnt/sde1/FPS/IdTech2/QuakeDev/id1/maps/test02_compile.bsp as Quake BSP bsp29

0.006s seconds elapsed

0 empty lightmaps

Finished with exit code 0

Executing '/mnt/sde1/FPS/IdTech2/QuakeDev/ironwail +game hellojam26map +map test02_compile +developer 1'

Command line: /mnt/sde1/FPS/IdTech2/QuakeDev/ironwail +game hellojam26map +map test02_compile +developer 1

Found SDL version 2.30.0

Terminal input not available.

Detected 8 CPUs.

Initializing Ironwail v0.8.2

Host_Init

Using add-on server "https://kexquake.s3.amazonaws.com"

Playing registered version.

Using renderer/network isolation.

Console initialized.

UDP Initialized

Server using protocol 999 (RMQ)

Exe: 22:41:55 Aug 11 2026 (Linux 64-bit)

384.0 megabyte heap

Video initialization

Video mode: 1280 x 636 24bit Z24 S8 60Hz

GL_VENDOR: NVIDIA Corporation

GL_RENDERER: NVIDIA GeForce RTX 3080/PCIe/SSE2

GL_VERSION: 4.3.0 NVIDIA 580.178.04

FOUND: EXT_texture_filter_anisotropic

FOUND: ARB_buffer_storage

FOUND: ARB_multi_bind

FOUND: ARB_bindless_texture

FOUND: ARB_shader_draw_parameters

FOUND: ARB_clip_control

Sound Initialization

SDL audio spec : 44100 Hz, 1024 samples, 2 channels

SDL audio driver: pulseaudio - X-USB Multichannel, 65536 bytes buffer

Audio: 16 bit, stereo, 44100 Hz

Language initialization

Loaded 1679 strings from 'localization/loc_english.txt'

========= Quake Initialized =========

Level editing environment detected, enabling map_checks

(pass -nomapchecks or set map_checks to 0 to disable)

Note: map_checks overrides gl_zfix, r_oit, and r_alphasort

Add-on server status:

 29 add-ons available for download

  0 add-ons already installed

execing quake.rc

execing default.cfg

execing ironwail.cfg

couldn't exec autoexec.cfg

Wrote ironwail.cfg.

"game" changed to "id1;hellojam26map"

Loaded 1679 strings from 'localization/loc_english.txt'

==========

Map checklist (test02_compile.bsp):

[x] lightmap data

[x] vis data

[ ] trigger_changelevel

[ ] info_intermission

[ ] skill spawnflags/triggers

[ ] info_player_coop (0/3+)

[ ] info_player_deathmatch (0/5+)

[ ] music track (worldspawn "sounds" field)

[ ] map title (worldspawn "message" field)

==========

3 demo(s) in loop

execing quake.rc

execing default.cfg

execing ironwail.cfg

FindFile: can't find autoexec.cfg

couldn't exec autoexec.cfg

"game" is already "id1;hellojam26map"

Sending clc_disconnect

SpawnServer: test02_compile

Clearing memory

Programs occupy 2403K.

trying to open external vis file

maps/test02_compile.vis not found, trying hellojam26map.vis

external vis not found

"mapversion" is not a field

[WORLDSPAWN] function started ...

[ENGINE] Advanced Engine

[WORLD] Reading SAVED1-4 console variables

[WORLD] Reading TEMP1 console variable (1024)

[WORLD] Max particles (512)

[PCHAIN] Creating controller

[WORLD] Switchable lights start (32)

[WORLD] Setup Light style array

[WORLD] Setup Pendulum vector array

[WORLD] Setup Map Var bitflag array

[WORLD] Setup Shadow lightstyle array

0 entities inhibited

Server spawned.

==========

Map checklist (test02_compile.bsp):

[x] lightmap data

[x] vis data

[ ] trigger_changelevel

[ ] info_intermission

[ ] skill spawnflags/triggers

[ ] info_player_coop (0/3+)

[ ] info_player_deathmatch (0/5+)

[ ] music track (worldspawn "sounds" field)

[ ] map title (worldspawn "message" field)

==========

CL_EstablishConnection: connected to local

changes will not take effect until the next level load.

Client LOCAL connected

[CLIENT] SetNewParms

[SVRFLAG] Info_Player_Start2 RESET

[SVRFLAG] Bef (0) BITs (112) Aft (0)

[SVRFLAG] Inventory detection RESET

[SVRFLAG] Bef (0) BITs (128) Aft (0)

[SetNewParms] RESETTING Map Variables

[SKILL4] EvilMode Disabled!

FITZQUAKE 0.85 SERVER (53716 CRC)

Serverinfo packet received.

Using protocol 999

Lightmap size: 256 x 256 (1/1 blocks)

Lightmap memory: 0.2 MB (5.9% efficiency)

FindFile: can't find music/track00.mp3

Couldn't find a cdrip for track 0

CL_SignonReply: 1

Programs occupy 94K.

Warning: [CL] check "FRIK_FILE" before calling strcat (csqc_hudad.qc: prior*CSQC_Init)

[RND_SYNC] Chaos Mode Disabled!

[PCHAIN] Part bank (0 / 512)

CL_SignonReply: 2

Warning: [SV] check "DP_QC_TOKENIZE_CONSOLE" before calling tokenize_console (csqc_hudad.qc: SV_ParseClientCommand)

player entered the game

[CLIENT] PutClientInServer

[SKILL4] EvilMode Disabled!

[COOP] Spawn locations Missing!

[CAMERA] Intermissions missing!

[CLIENT] Decode (Level) Parameters

[CLIENT] Decode Worldspawn Parameters!

[LevelParms] Reading Map Variables

[RND_LOAD] Checking load exceptions!

[CONFIG] MOD Settings

- Particle System disabled

+ Engine selects particle system

- Weather System disabled

+ Client Server QC HUD enabled

+ Scratch1 CSQC Hud (5) active

+ Shotgun projectiles system

+ Shotgun casing enabled

+ Large Projectile Collision

+ MWheel ng/sng skip enabled

+ Z Aware Monster combat enabled

+ footstep sounds enabled

+ monster liquid damage enabled

+ Dev helpers (mark/arrow) enabled

- All items use corner origin

+ All items are randomly rotated

+ Player weapon autoaim disabled

+ Arcane Key Inventory Msg enabled

+ No fade away for monster bodies

- No floor checks for monster bodies

+ Missile tracking uses default

* Global fog default not defined

* Sky fog default not defined

+ Lightstyle compiler lights at (32)

+ DP features ( )

CL_SignonReply: 3

368.5 megabyte data cache

[INFO] Arcane Dimensions V1.90 (Beta4)

CL_SignonReply: 4

Warning: [CL] check "DP_QC_SINCOSSQRTPOW" before calling pow (csqc_hudad.qc: CSQC_DrawHud)

Warning: [CL] check "FRIK_FILE" before calling strlen (csqc_hudad.qc: Hud_DrawNoFont24)

Warning: [CL] check "FTE_STRINGS" before calling str2chr (csqc_hudad.qc: Hud_DrawNoFont24)

[PCHAIN] Part bank (64 / 512)

[PCHAIN] Part bank (128 / 512)

[PCHAIN] Part bank (192 / 512)

[PCHAIN] Part bank (256 / 512)

[PCHAIN] Part bank (320 / 512)

[PCHAIN] Part bank (384 / 512)

[PCHAIN] Part bank (448 / 512)

Sending clc_disconnect

Client player removed

Wrote ironwail.cfg.

Shutting down SDL sound

Finished with exit code 0

Worked for 18s

ChatGPT said:

You said:

Its all there:
https://github.com/ChnuckiErdbeer/Quake-Surface-Studio
there should be 3 folders:
"quake_surface_studio (zip me for blender)" contains the extracted zip with all the source and tests
"blenderfile" containst a single .blend file i used to test the addon
"mapandbsp" a .map file a.bsp file and all files created alongside the .bsp creation.
can you really not see this files?

Worked for 1m 51s

ChatGPT said:

You said:

very good.

Worked for 3m 5s

ChatGPT said:

You said:

close :D you are right that 8 was the wrong entity count. But 10 is also wrong. ich has 9: six containing delay values 0 - 5 and three with different color values. 40 face sounds right. The original map file consists of 6 cubuids formimg the room (four walls, ceiling and floor) and a sixth forming the pillar. This originally makes 42 brushes but since the compiling process removes non visible faces and splits some for vis / light optimization the bsp face numer can differ. Everything seems fine. Keep on developing until you need me to test somethin or have questions.

Worked for 5m 41s

ChatGPT said:

You said:

Imported 40 faces, 10 light entities. Palette: /mnt/sde1/FPS/IdTech2/QuakeDev/id1/gfx/palette.lmp. Textures: 4 embedded, 0 WAD, 0 external, 0 white fallback. RGB lightmaps: QLIT v1 .lit sidecar. Spawn view set to info_player_start.
bpy.ops.object.editmode_toggle()
bpy.ops.mesh.select_mode(use_extend=False, use_expand=False, type='FACE')
bpy.ops.object.editmode_toggle()
bpy.ops.object.editmode_toggle()
bpy.ops.qss.detach_selected_faces()
Detached 1 face(s) into separate 0–1 paint textures. Use 'Enter Texture Paint' to paint the active selected face.
bpy.ops.object.editmode_toggle()
bpy.ops.paint.texture_paint_toggle()
Painting image: QSS_Face_00037_Paint

ChatGPT said:

You said:

painting the face in paint mode does not leave any visibnle marks. seems to not work or paint on a not visible texture "layer"

Worked for 1m 48s

ChatGPT said:

You said:

it works now.. kind of:
the paint appears but strangley offset from my mouse cursor. on the detached face tho.
Heres the output:
bpy.ops.object.delete(use_global=False, confirm=False)
Deleted 1 object(s)
bpy.context.space_data.system_bookmarks_active = 3
Modules Installed (quake_surface_studio) from '/home/chnucki/Downloads/quake_surface_studiofix/v0.11.1-metadata-style-order-test.zip' into '/home/chnucki/.var/app/org.blender.Blender/config/blender/5.2/scripts/addons'
bpy.context.region.active_panel_category = 'Quake'
bpy.context.space_data.recent_folders_active = 2
bpy.data.screens["temp.002"]. = "QuakeDev"
bpy.context.space_data.recent_folders_active = 2
Imported 40 faces, 10 light entities. Palette: /mnt/sde1/FPS/IdTech2/QuakeDev/id1/gfx/palette.lmp. Textures: 4 embedded, 0 WAD, 0 external, 0 white fallback. RGB lightmaps: QLIT v1 .lit sidecar. Spawn view set to info_player_start.
bpy.context.space_data.shading.type = 'RENDERED'
bpy.ops.object.editmode_toggle()
bpy.ops.mesh.select_mode(use_extend=False, use_expand=False, type='FACE')
bpy.ops.object.editmode_toggle()
bpy.ops.object.editmode_toggle()
bpy.ops.qss.detach_selected_faces()
Detached 1 face(s) into separate 0–1 paint textures. Use 'Enter Texture Paint' to paint the active selected face.
bpy.ops.object.editmode_toggle()
bpy.ops.paint.texture_paint_toggle()
Painting image (Single Image canvas): QSS_Face_00037_Paint

I guess you can also work in the main branch or at least give me some versioned release zips again so i dont have to puzzle them togeher by hand for blender.

Worked for 3m 55s

ChatGPT said:

You said:

Ok something is off. I detach a face, click enter texture paint, the paint appears on the faces left to the one detached. Then i detached another one and the paint is nowhere to be found.

Worked for 3m 25s

ChatGPT said: